

Statistical Analysis

1. Load libraries “ggplot2” and “dplyr”
2. Read in my main cleaned dataset
3. Collect a general statistical summary of monthly percent change
 - a. Minimum -44% change
 - i. COVID-19 recovery period
 - b. Maximum 348% change
 - i. COVID-19 jump
4. Read in my cleaned population dataset
5. Define the states to be used
 - a. Arizona
 - b. Arkansas
 - c. New Jersey
 - d. Maine
 - e. Tennessee
 - f. California
6. Filter for the years that overlap between the datasets
 - a. 2020-2025
7. Join the datasets by year and state
8. Transform monthly values into yearly values
9. Ensure the necessary variables are integers
10. Filter the years down again
11. Merge the data
12. Graph a distribution of population density
 - a. Heavily skewed right
 - i. Correlation test may need to be logarithmically altered
 - b. One state has very very high population density
13. Pearson Correlation Test (Population density and unemployment)
 - a. P-value: 0.0946
 - i. Not statistically significant, outside of the 95% CI
 - ii. Implies a lack of relationship between my variables
 - b. Correlation coefficient: 0.2828599
 - i. Low, but positive
 - ii. Correlation unlikely

14. Analyze Pearson test per state
 - a. Only California has a positive correlation
 - b. Population density is likely an irrelevant factor
 - c. Denser state, higher unemployment- However, unemployment will fall while density is consistent
 - d. Generally above 50%, either way positive or negative- Nothing above ~81
15. Retest Pearson with a logarithmic transformation
 - a. P-value: 0.02295
 - i. Statistically significant
 - ii. Better than the prior, within the 95% CI
 1. There is some correlation between population density and unemployment
 - b. Correlation coefficient: 0.3781806
 - i. Still low and still positive
 - ii. Better than the previous
16. Plot Log Pearson results
 - a. Linear line to compare different relationships
 - i. Points collected around left, middle, and right
 - ii. Generally positive
17. Read in dataset of all states
18. Set up all states data to be merged
 - a. Group by state, year
19. Join all states data to population data
20. Double check that everything is functioning as intended
21. Pearson's correlation test between all states unemployment and population density
 - a. p-value: 0.22
 - i. Not statistically significant
 - b. Correlation coefficient: 0.10
 - i. Positive
 - ii. Very low
22. Graph histogram of population density distribution
 - a. Heavily right skewed
 - i. Consider log transformation
23. Log transform unemployment, population density

24. Retest pearson's correlation test

- a. p-value: $7.13e-11$ (Nearly zero)
 - i. Statistically significant
- b. Correlation coefficient: 0.50
 - i. Positive
 - ii. Much higher

25. Graph log transformed variables with linear line to visualize correlation

- a. Notable positive trend with point clustering
- b. More spread out points